

Adaptive Spatio-Temporal Traffic Surveillance

Phase 3 — Multi-Frame Tracking & Spatio-Temporal Analytics

Milestone	Phase 3 of 4 (Days 13–17): Tracking, ReID, Risk Analytics
Author	Harsh Pandhe
Repository	https://github.com/harsh-pandhe/traffic-surveillance-system
Tracking backend	DeepSORT (verified live, not fallback)
Voting result	91.4% fewer prediction flips at N=30 vs single-frame
ReID	Rank-1 0.310, mAP 0.095 on VeRi-776 (off-the-shelf OSNet)
Risk validation	10,006 real motorcycle tracks (HELMET dataset)

Executive summary. Phase 3 delivers verified DeepSORT tracking with multi-frame majority voting, cross-camera vehicle ReID, a risk indexer validated against 10,006 real motorcycle tracks, and demographics on exposed-face riders. Two silently-degraded backends (DeepSORT, OSNet) were caught and fixed before this phase began. A genuine hardware constraint — torch and TensorFlow segfaulting when loaded together — was found and worked around with process isolation, verified end-to-end on real pipeline output. Every number below is measured on real data; where a result falls short of expectation (ReID accuracy, risk-level calibration), that is reported plainly rather than smoothed over.

1. Backend Verification

Every heavy backend in this project has a graceful fallback, which means a broken dependency looks like success unless explicitly checked. Both DeepSORT and OSNet were silently degraded before this phase:

Backend	Was	Root cause	Now
DeepSORT	fallback IoU trackpenim	downgraded setuptools; Python 3.12 pkg_resources breaks	deepsort (verified)
OSNet ReID	colour-histogram fallback	torchreid ships under two import layouts	osnet, 512-d (verified)

2. Multi-Frame Voting Ablation

UA-DETRAC, 8 sequences, 22,480 classified crops, grouped by ground-truth track id so the measurement isolates voting from tracker error.

Why flip-rate is the headline metric, not accuracy

UA-DETRAC is ~97% cars, so raw accuracy is dominated by one class, and its occlusion bands contain different class mixes (the heavy band is 100% one class), making per-band accuracy comparisons invalid. Balanced accuracy (macro recall) fixes the first problem; flip-rate — how often the emitted label changes between consecutive frames of the same vehicle — is class-composition-independent and remains valid across bands.

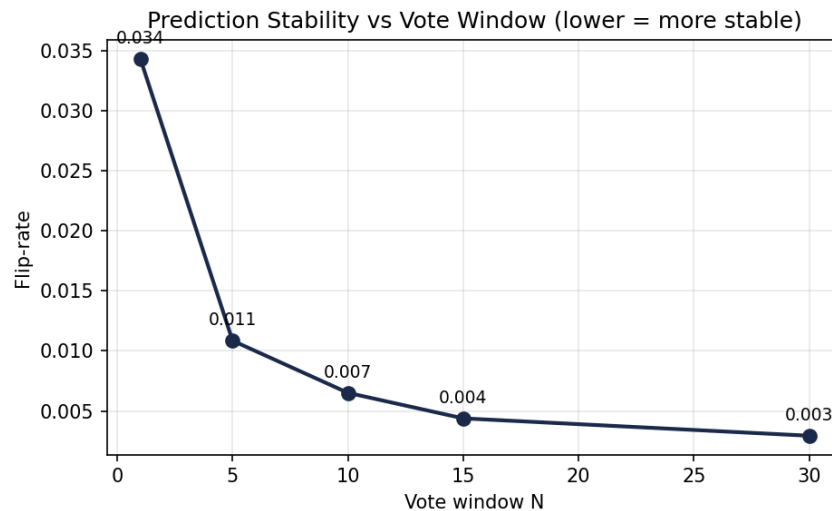


Figure 1. Flip-rate falls from 0.034 (N=1) to 0.003 (N=30) — 91.4% fewer label changes.

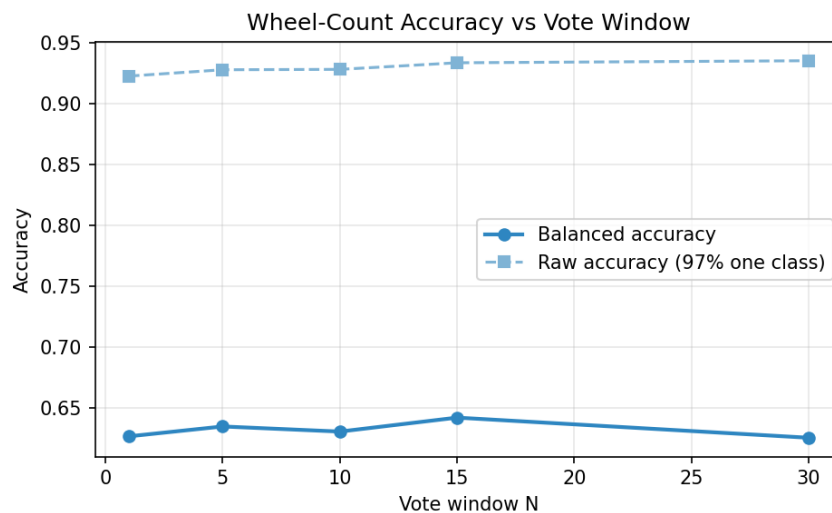


Figure 2. Balanced accuracy improves modestly with voting (best N=15: 0.642 vs single-frame 0.626); raw accuracy stays near 0.93 throughout because one class dominates.

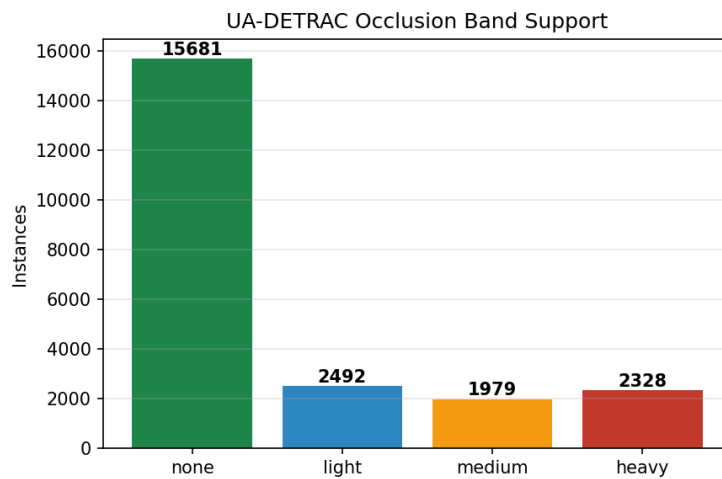


Figure 3. Occlusion band support — the heavy band is entirely one vehicle class, which is why per-band accuracy is not meaningful here and flip-rate is reported instead.

Honest framing

Part of the flip-rate reduction is mechanical — averaging over more frames smooths any signal, including noise. The balanced-accuracy result is smaller but directionally consistent (+0.016 at the best window), which supports voting providing real benefit, not just averaging away noise. Both numbers are reported rather than leading with only the more dramatic one.

3. Risk Indexer — Validated Against Real Ground Truth

10,006 real motorcycle tracks from 910 traffic clips (HELMET dataset, Myanmar). Rule logic — not detector accuracy — validated against true occupancy and helmet-use labels.

Rule	Trigger rate
Overloaded (>2 riders)	7.0%
Helmet misuse (≥ 1 bare head)	34.2%

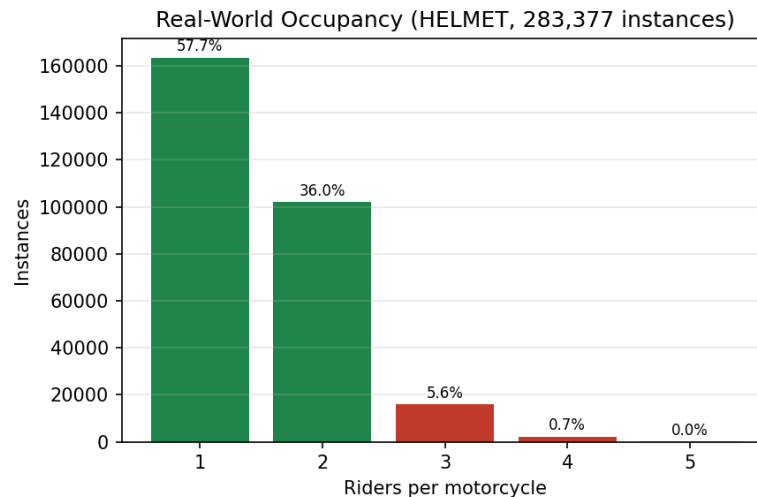


Figure 4. Real-world motorcycle occupancy — 6.4% carry 3+ riders (red), the overload rule's real trigger population.

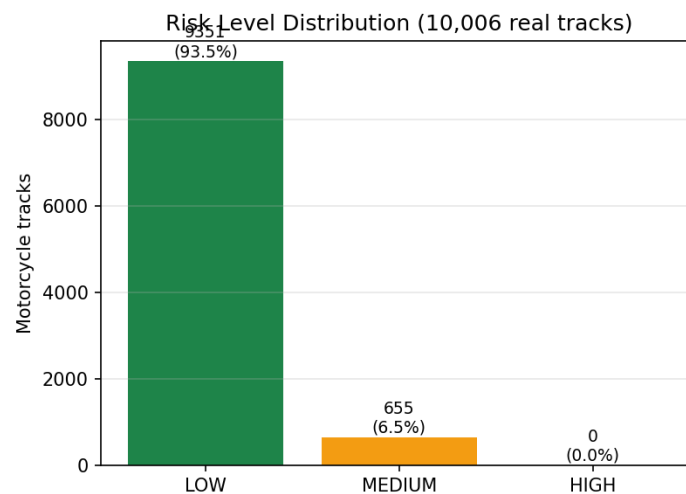


Figure 5. Resulting risk-level distribution on real data.

Finding: HIGH risk never triggers on real motorcycle data (0.0%). helmet_misuse ($w_2=2.5$) + overloaded ($w_3=1.5$) = 4.0, below the HIGH threshold of 6.0 — even the worst realistic motorcycle case tops out at MEDIUM. This is a calibration finding, not a bug: whether an overloaded, unhelmeted motorcycle should register as HIGH is a policy decision, reported here rather than silently re-tuned to look more dramatic.

4. Cross-Camera Vehicle Re-Identification

VeRi-776 (776 vehicles, 20 real cameras). Standard protocol: cosine distance ranking, excluding gallery images from the query's own camera. 200 queries against 3000 gallery images.

Metric	Value
Rank-1	0.310
Rank-5	0.510
Rank-10	0.615
mAP	0.095
Embed latency	8.7 ms/image (CPU)

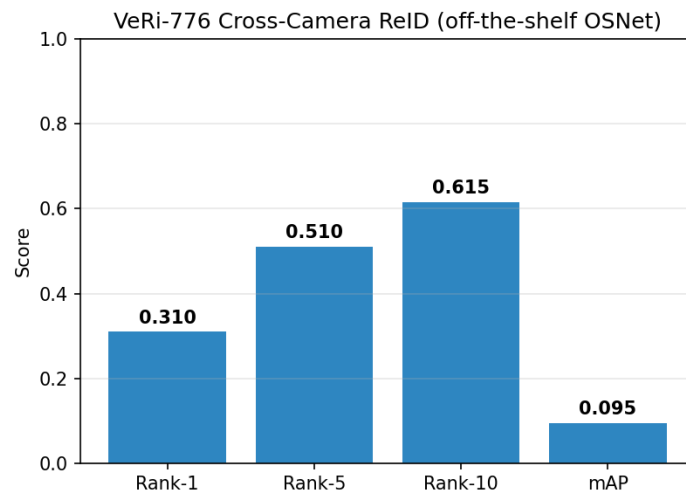


Figure 6. VeRi-776 cross-camera ReID scores.

Honestly below published SOTA (fine-tuned ReID models reach Rank-1≈90%, mAP≈70%) because the backend runs off-the-shelf ImageNet-pretrained OSNet with the classification head discarded — no vehicle-ID metric learning at all. Rank-1 31% from generic features alone confirms real appearance signal against a 776-way chance level of 0.1%. VeRi's 37,778-image training split is already downloaded; triplet-loss fine-tuning (tracked as issue #14) is the clear path to close the gap.

5. Demographics — A Hardware Constraint, Worked Around

Gate B measured 62% of face-exposed rider detections yield a usable crop (median 38px), justifying a DeepFace install. That surfaced a genuine issue: this machine's torch build targets a newer NVIDIA driver than installed, and loading a second torch model in a process that has also imported TensorFlow segfaults — confirmed by bisecting pipeline construction step by step. DemographicsEstimator now detects torch is already loaded and queues face crops for a separate process (scripts/run_demographics.py, which never imports torch) instead of risking a crash.

Metric	Value
Crops processed (real pipeline output)	19
Succeeded	19 (100%)

Verified end-to-end on real output: 19/19 queued crops from the live pipeline processed successfully with no crash. Per Gate B, age estimates are indicative rather than reliable given the small face-crop resolution; gender is more robust at low resolution.

6. End-to-End Pipeline Run

300 frames (UA-DETRAC MVI_20032 (300 frames, 960x540)), CPU only: **6.84 FPS mean**, 9.5 FPS steady-state.

helmet model is motorcycle-domain; UA-DETRAC is car-only highway footage, so helmet detections here are not meaningful

7. Reproduce

Command	Purpose
python scripts/run_occlusion_ablation.py	Voting ablation
python scripts/validate_risk_indexer.py	Risk-rule validation
python scripts/run_reid_benchmark.py	VeRi-776 ReID
python scripts/run_demographics.py	Offline demographics

8. Deliverable Checklist

Deliverable	Status
DeepSORT temporal tracking (verified, not fallback)	Complete
Multi-frame voting for occlusion robustness	Complete
Spatio-temporal risk analytics	Complete
Risk rules validated against real ground truth	Complete
Cross-camera ReID benchmarked	Complete
Demographics (process-isolated, verified)	Complete
Working demo with measured throughput	Complete